Raffles Institution
Year 5-6 Physics Department

2022 Year 6 H2 Physics Term 2 Timed Practice Solution and Mark Scheme

Part 1

1 **(a)** Newton's law of gravitation states that the force of attraction F between two point masses m and M, separated by a distance R is given by $F = \dfrac{GMm}{R^2}$ where G is a constant of proportionality.

Since gravitational field strength g at a point is defined as the gravitational force experienced per unit mass at that point that is at a distance R from M

$$\begin{aligned} g &= \frac{F}{m} \\ &= \frac{GMm}{R^2} \div m \\ &= \frac{GM}{R^2} \end{aligned}$$

*Need to define gravitational field strength in words as the question stated to use the definition.
*Use symbols given as defined in the question. Other symbols used have to be defined.

(b) **(i)**

$$\frac{g_{\text{neutron star}}}{g_{\text{sun}}} = \left(\frac{M_{star}}{M_{sun}}\right)\left(\frac{R_{sun}}{R_{star}}\right)^2$$

$$\frac{g_{\text{neutron star}}}{290} = (1.5)\left(\frac{1}{10^{-5}}\right)^2$$

$$g_{\text{neutron star}} = 4.35 \times 10^{12} \text{ N kg}^{-1}$$

(ii) **1.**

$$\begin{aligned} a &= r\omega^2 \\ &= \left(1.1 \times 10^4\right)\left(\frac{2\pi}{0.24}\right)^2 \\ &= 7.5392 \times 10^6 = 7.54 \times 10^6 \text{ m s}^{-2} \end{aligned}$$

2. The gravitational field strength at the surface of the star is much larger than the required centripetal acceleration to keep the particles in circular motion.

OR

Since the gravitional field strength at the surface of the star is much larger than the centripetal acceleration at the surface of the star, this implies that there is normal contact force on the particles at the surface.

Hence, particles on the surface of the star will not leave the surface.

(iii) The gravitational field strength on the surface of a neutron star is 10^{12} times larger than that of earth's. Such an immense gravitational field strength pulls everything on the surface inwards and does not allow anything tall to exist.

2 **(a)**

$$I_0 = \frac{k}{5.0^2} \quad \text{----- (1)}$$

$$\frac{I_0}{3} = \frac{k}{(5.0+x)^2} \quad \text{----- (2)}$$

$$\frac{(1)}{(2)}: \quad 3 = \frac{(5.0+x)^2}{5.0^2}$$

$$(5.0+x)^2 = 75$$

$$5.0 + x = \sqrt{75}$$

$$x = 3.6603 = 3.66\text{ m}$$

(b) **(i)** If the light from the light source was unpolarised, the polarising filter would have caused the intensity measured to be reduced to $\frac{I_0}{2}$ and not $\frac{I_0}{5}$.

(ii) Using Malus's Law,

$$\frac{I_0}{5} = I_0 \cos^2\theta$$

$$\cos^2\theta = \frac{1}{5}$$

$$\theta = 63.435 = 63.4°$$

(c) The reflector directs the light waves towards the detector which increases the power of the light wave arriving at the detector but spread over the same area of the detector. Since the intensity of a wave is the power per unit area normal to the direction of energy transfer, I_1 is larger than I_0.

OR

The reflector directs the light waves towards the detector such that the same power emitted from the light source is now spread over a smaller wavefront area at the detector. Since the intensity of a wave is the power per unit area normal to the direction of energy transfer, I_1 is larger than I_0.

3 **(a)** The first law of thermodynamics states that the increase in the internal energy of a system is equal to the sum of the heat supplied to the system and the work done on the system, and the internal energy of a system depends only on its state.

(b) **(i)** For ideal gas, $pV = nRT$. Since n and R are constants,

$$\frac{p_A V_A}{T_A} = \frac{p_B V_B}{T_B}$$

$$V_A = \left(\frac{T_A}{T_B}\right)\left(\frac{p_B}{p_A}\right)(V_B)$$

$$= \left(\frac{130}{500}\right)\left(\frac{8.3\times10^5}{3.7\times10^5}\right)\left(6.4\times10^{-3}\right)$$

$$= 3.733\times10^{-3} = 3.7\times10^{-3}\ \text{m}^3 \ \text{(shown)}$$

*Write the non-rounded off value (at least 1 more s.f. than the given value) before writing the rounded off value as given in the question.

(ii) Since the internal energy of an ideal gas is directly proportional to its temperature, there is an increase in internal energy from A to B as temperature increases. Hence $\Delta U > 0$.

Since there is an increase in volume, work done on the gas is negative. Hence $W < 0$.

By the first law of thermodynamics, heat supplied to the gas $Q = \Delta U - W$ is positive. Hence heat is supplied to the gas.

(iii)

$$U = \frac{3}{2}pV$$

$$U_C - U_B = \frac{3}{2}(\Delta p)V$$

$$= \frac{3}{2}\left[(3.7-8.3)\times10^5\right]\left(6.4\times10^{-3}\right)$$

$$= -4416 = -4420\ \text{J}$$

(iv) In the cyclic process ABCA, $\Delta U_{net} = 0$

net work done on gas, $W_{net} = -$area enclosed

$$W_{net} = -\frac{1}{2}\left[(8.3-3.7)\times10^5\right]\left[(6.4-3.7)\times10^{-3}\right] = -621\ \text{J}$$

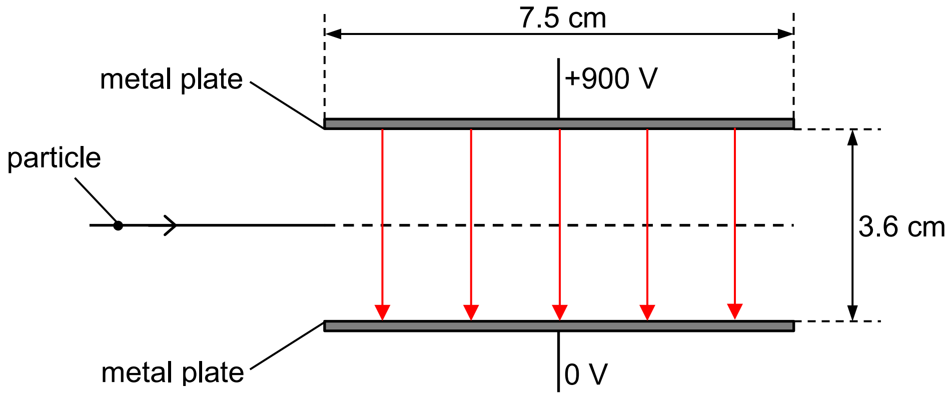
$$Q_{net} = \Delta U_{net} - W_{net}$$

$$= 0-(-621)$$

$$= 621\ \text{J}$$

4 **(a)** The electric field strength is the same at all points within the field.
OR
The electric potential gradient is constant.
OR
The change in electric potential per unit distance is the same everywhere within the field.
OR
The force per unit charge is the same everywhere within the field.
OR
The same charge experiences the same force everywhere within the field.

(b) **(i)**



*At least 5 parallel lines, equally spaced, pointing downwards

(ii) Points vertically upwards.
*It is important to include 'vertically'. Just 'towards' or 'up' is vague.

(c) **(i)** $F = ma$

$$a = \frac{F}{m} = \frac{Eq}{m} = \left|\frac{\Delta V}{d}\right|\frac{q}{m}$$

$$= \frac{(900)\left(2 \times 1.60 \times 10^{-19}\right)}{\left(3.6 \times 10^{-2}\right)\left(6.6 \times 10^{-27}\right)}$$

$$= 1.2121 \times 10^{12} = 1.21 \times 10^{12} \text{ m s}^{-2} \quad \text{(shown)}$$

*Write the non-rounded off value before writing the rounded off value given.

(ii) Consider horizontal motion,
time taken for particle to travel through the plates

$$s_x = u_x t + \frac{1}{2}a_x t^2 \qquad \text{where } a_x = 0$$

$$t = \frac{s_x}{u_x} = \frac{7.5 \times 10^{-2}}{4.1 \times 10^5} = 1.8292 \times 10^{-7} \text{ s}$$

Consider vertical motion,
vertical displacement in the same time

$$s_y = u_y t + \frac{1}{2}a_y t^2 \qquad \text{where } u_y = 0$$

$$s_y = \frac{1}{2}\left(1.21 \times 10^{12}\right)\left(\frac{7.5 \times 10^{-2}}{4.1 \times 10^5}\right)^2 = 0.02024 = 0.0202 \text{ m}$$

maximum possible vertical displacement between the plates is $\frac{0.036}{2} = 0.018$ m

Since $\underline{0.0202 \text{ m} > 0.018 \text{ m}}$, the particle collides with the upper plate.

OR

Consider vertical motion,
time taken for particle to travel the vertical displacement to plate,

$$s_y = u_y t + \frac{1}{2}a_y t^2 \quad \text{where } u_y = 0$$

$$\frac{0.036}{2} = \frac{1}{2}\left(1.21\times10^{12}\right)t^2$$

$$t = \sqrt{\frac{0.036}{1.21\times10^{12}}} = 1.7248\times10^{-7} \text{ s}$$

Consider horizonal motion,
horizontal displacement in the same time,

$$s_x = u_x t + \frac{1}{2}a_x t^2 \quad \text{where } a_x = 0$$

$$s_x = \left(4.1\times10^5\right)\left(\sqrt{\frac{0.036}{1.21\times10^{12}}}\right) = 0.07072 = 0.0707 \text{ m}$$

Since $\underline{0.0707 \text{ m} < 0.075 \text{ m}}$, the particle collides with the upper plate.

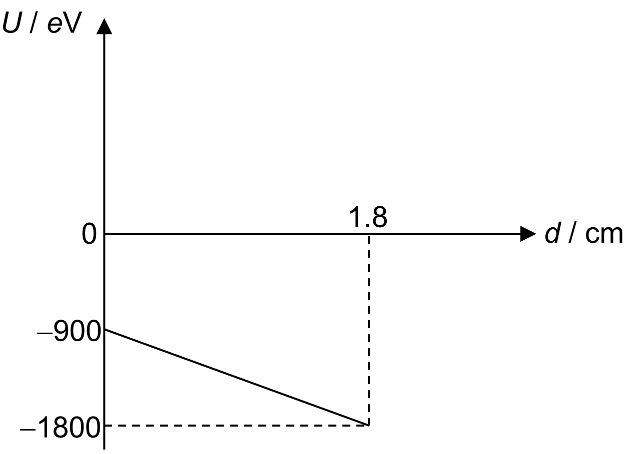
(iii) Since the potential V between the plates is positive everywhere and the particle is negatively charged,
potential energy, $U = qV = -2eV$ is negative everywhere.

Since electric field is uniform, potential and hence potential energy varies linearly with d.

At point of entry, $U = -2eV = -2e\left(\frac{900}{2}\right) = -900$ eV

When particle hits the upper plate, $U = -2eV = -2e(900) = -1800$ eV

For the particle, potential energy decreases, kinetic energy increases.



5 **(a)** **(i)**

$$V = IR = \frac{E}{R+r}R$$
$$1.20 = \frac{1.50}{R+0.162}R$$
$$R + 0.162 = \frac{1.50}{1.20}R$$
$$\frac{1.50}{1.20}R - R = 0.162$$
$$R = 0.648\ \Omega \text{ (shown)}$$

OR

$$\frac{R}{R+r} = \frac{V_\mathrm{R}}{E}$$
$$R = \frac{V_\mathrm{R}}{E}\left(R+r\right)$$
$$R = \frac{1.20}{1.50}R + \frac{1.20}{1.50}r$$
$$R - \frac{1.20}{1.50}R = \frac{1.20}{1.50}\left(0.162\right)$$
$$R = 0.648\ \Omega \text{ (shown)}$$

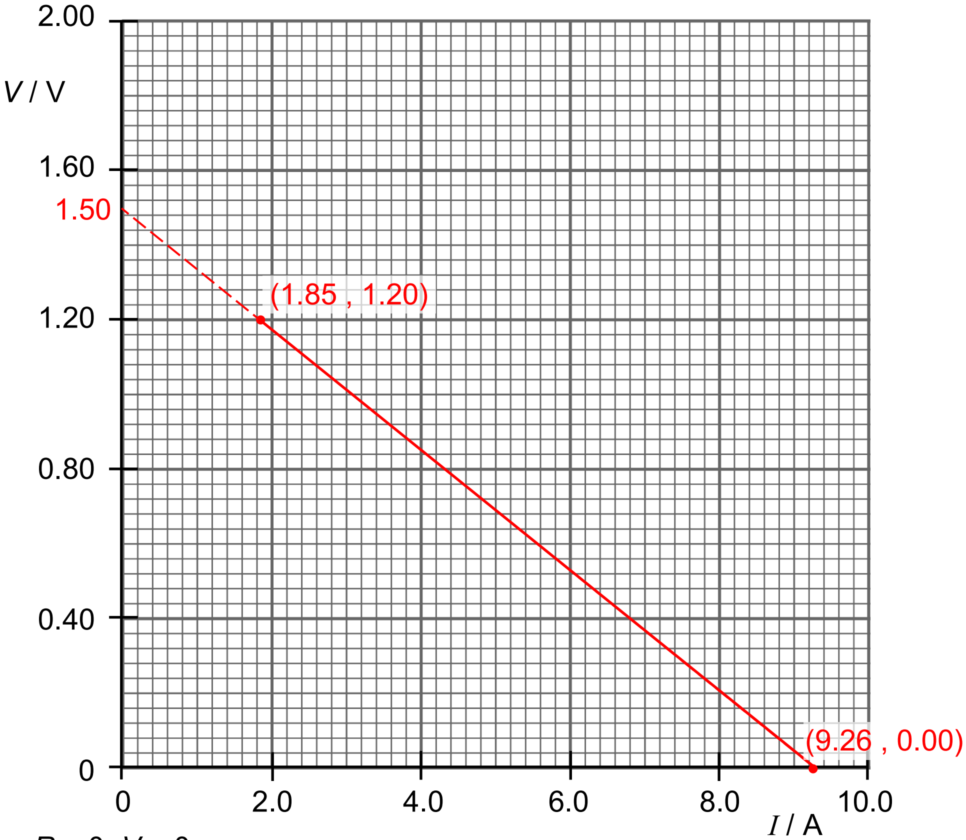
(ii)

$$I = \frac{V}{R}$$
$$= \frac{1.20}{0.648}$$
$$= 1.852 = 1.85\ \text{A}$$

OR

$$I = \frac{E}{R_\text{total}}$$
$$= \frac{1.50}{0.648+0.162}$$
$$= 1.852 = 1.85\ \text{A}$$

(iii)



when $R = 0$, $V = 0$

$$I = \frac{E}{r} = \frac{1.50}{0.162} = 9.259 \text{ A}$$

(iv) The non-ideal ammeter has resistance R_{ammeter}.

$$V = E - Ir \quad \text{where} \quad I = \left(\frac{E}{r + R_{\text{ammeter}} + R}\right)$$

Similarities:

1. The equation $V = E - Ir$ still applies. Hence gradient remains as $-r$.
2. The vertical intercept remains as E.

Difference:

1. The coordinates of the start and end points of the graph. The resistance of the ammeter adds to the total resistance of the circuit, which results in different I and V values at each value of R. However, these two points will still lie along the same line drawn in (a)(iii).

(b) **(i)** P is maximum when $R = r$

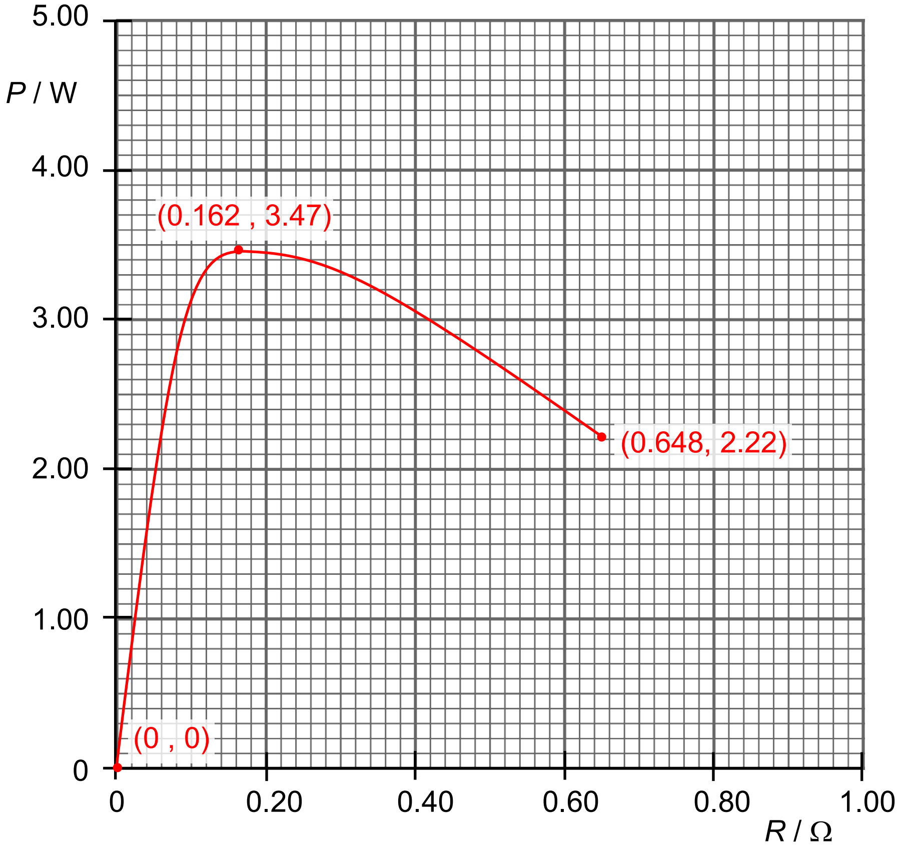
$$P_{\max} = I^2 R$$

$$= \left(\frac{E}{r + r}\right)^2 r$$

$$= \left(\frac{1.50}{2 \times 0.162}\right)^2 (0.162)$$

$$= 3.472 = 3.47 \text{ W}$$

(ii)



$$P = I^2R = \left(\frac{E}{R+r}\right)^2 R$$

At $R = 0\ \Omega$, $P_0 = \left(\dfrac{1.50}{0+0.162}\right)^2 (0) = 0$ W

At $R = r$, $P_{\max} = \left(\dfrac{1.50}{0.162+0.162}\right)^2 (0.162) = 3.47$ W

At $R = 0.648\ \Omega$, $P_{0.648} = \left(\dfrac{1.50}{0.648+0.162}\right)^2 (0.648) = 2.22$ W

Part 2

6 **(a)** **(i)** Since k, E, L and M are constants, $a \propto -x$.

The acceleration a of the strip is directly proportional to its displacement x from the equilibrium position.

The negative sign implies that the acceleration a acts in the opposite direction to the displacement x from the equilibrium position. Hence acceleration acts towards the equilibrium position.

These satisfy the conditions for simple harmonic motion.

(ii) **1.** initial depression = 3.0 cm

2. amplitude = 1.2 cm

3. $\omega = 2\pi f = 34.9$

$f = \dfrac{34.9}{2\pi} = 5.5545 = 5.55$ Hz

4. Comparing SHM equation $a = -\omega^2 x$ with $a = -\dfrac{kE}{L^3 M} x$

$$\omega^2 = \frac{kE}{L^3 M}$$

$$k = \frac{\omega^2 L^3 M}{E}$$

$$= \frac{34.9^2 \times 0.60^3 \times 0.12}{7.0 \times 10^{10}}$$

$$= 4.5101 \times 10^{-10} = 4.51 \times 10^{-10}$$

(b) **(i)** **1.** A forced oscillation occurs when a body is subjected to a periodic external driving force and is made to oscillate at the frequency of the driving force, which may not be its natural frequency.

2. Resonance

(ii) **1.** At max. amplitude, $f = 11.8$ Hz and $x_0 = 4.25$ cm

$$v_0 = \omega x_0$$

$$= 2\pi f x_0$$

$$= 2\pi(11.8)(4.25 \times 10^{-2})$$

$$= 3.1510 = 3.15 \text{ m s}^{-1}$$

2. $|a_0| = \omega^2 x_0$

$$= (2\pi f)^2 x_0$$

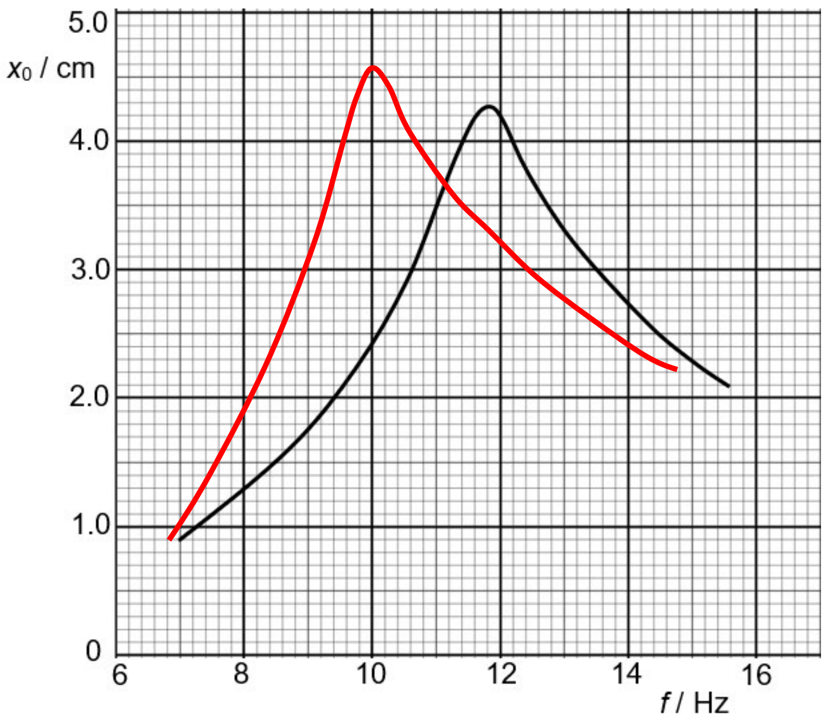
$$= [2\pi(11.8)]^2 (4.25 \times 10^{-2})$$

$$= 233.62 = 234 \text{ m s}^{-2}$$

3. max. speed occurs when body is at the equilibrium position
max. acceleration occurs when body is at maximum displacement
time interval between a max. speed and subsequent max. acceleration

$$= \frac{1}{4}T$$
$$= \frac{1}{4}\left(\frac{1}{f}\right)$$
$$= \frac{1}{4 \times 11.8}$$
$$= 0.021186 = 0.0212 \text{ s}$$

(iii)



Using springs of a smaller force constant will result in a lower natural frequency and a larger amplitude of oscillation at resonance.

For spring mass system, $T = 2\pi\sqrt{\frac{m}{k}} \quad \Rightarrow \quad f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

Hence, smaller k, smaller f.

Total energy, $E = \frac{1}{2}m\omega^2 x_o^{\ 2}$ remains constant

Hence smaller f, smaller ω, larger x_0.

7 **(a)** The principle of superposition states that when two or more waves of the same kind meet at a point, the resultant displacement at that point is equal to the vector sum of the displacements of the individual waves at that point.

(b) **(i)** The light waves entering each slit will diffract and overlap / meet. They will interfere according to the principle of superposition.

At locations along the screen where the path difference between the two waves is an integer multiple of the wavelength, the two waves meet in phase and undergo constructive interference, producing fringes of high intensity.

At locations along the screen where the path difference between the two waves is an odd integer multiple of half-wavelength, the two waves meet π radians out of phase and undergo destructive interference, producing fringes of low intensity.

*If used $n\lambda$ for path difference, *n* must be clearly defined as an integer.

(ii) **1.** The two light waves have a constant phase difference.

2. The two light waves must have similar amplitude.
OR
The two light waves must be unpolarised or polarised in the same plane.

(iii) Fringe separation

$$x = \frac{\lambda D}{a}$$
$$= \frac{(589 \times 10^{-9})(2.3)}{1.20 \times 10^{-3}}$$
$$= 1.1289 \times 10^{-3} = 1.13 \times 10^{-3} \text{ m}$$

(iv) **1.** Missing orders of the double slit interference pattern are those that coincide with the single slit diffraction minimas.

Minima of single slit diffraction pattern
$b \sin\theta = m\lambda$ ----- (1)
Maxima of double slit interference pattern
$a \sin\theta = n\lambda$ ----- (2)

$\frac{(2)}{(1)}$: $\frac{n}{m} = \frac{a}{b} = \frac{1.2}{0.30} = 4$
when $m = 1$, $n = 4$
when $m = 2$, $n = 8$

1st minimum and 2nd minimum of single slit diffraction pattern coincide with 4th order maximum and 8th order maximum of double slit interference pattern respectively. Hence, 4th and 8th order maxima of double slit interference pattern will be missing.

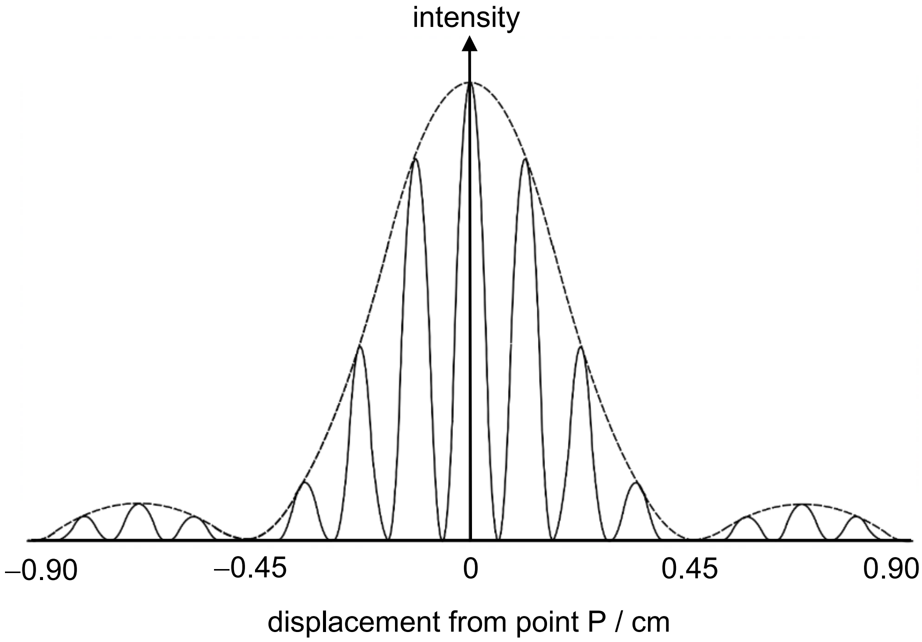
2. Distance from P to 4th maximum $= 4\left(1.13\times10^{-3}\right) = 0.00452\text{ m} = 0.452\text{ cm}$

Distance from P to 8th maximum $= 8\left(1.13\times10^{-3}\right) = 0.00904\text{ m} = 0.904\text{ cm}$

*Draw single slit diffraction pattern.
Width of the central envelope is twice the width of the neighbouring envelopes.
Intensity of the central envelope is much higher than the neighbouring envelopes.

*Double slit interference pattern with equal spacing between maximas.
Correct missing orders (4th and 8th).

*Ensure that pattern is symmetrical about P and has no sharp turning points.



(v) **1.** $d\sin\theta = n\lambda$

$$\sin\theta = \frac{n\lambda}{d}$$

$$\sin\theta \le 1$$

$$\frac{n\lambda}{d} \le 1$$

$$n \le \frac{d}{\lambda} = \frac{10^{-3}/500}{589\times10^{-9}} = 3.396$$

Number of orders on each side is 3.

2. A diffraction grating has more slits hence allows more light waves to pass through. Hence the maximum intensity is higher as it is formed from the constructive interference of more light waves.

Each bright fringe occupies a smaller area which also results in higher intensity light.